Daniele Battesimo Provenzano

@ daniele.provenzano@sns.it | Personal Website

Employment

Italian Association for the Teaching of Physics (AIF) — Various locations across Italy
Lecturer, Coach and Examiner — *May 2019 – present*

- Preparing and delivering lectures, lab sessions, and problem-solving activities for students training for the Italian, European and International Physics Olympiads.
- Writing exam and competition problems; contributing to national programs that promote physics education and support talented students across Italy.

Forensic Department of the Italian Police — Rome, Italy
Technical Inspector in Forensic Science and Lecturer — *Sept. 2025 – present*

- SEM and microanalytical investigations in the GSR Analysis Division.
- Teaching within internal training programs for forensics trainees.

Education

Scuola Normale Superiore — Pisa, Italy
Ph.D. in Physics (plasmonics and metamaterials); ***cum Laude*** — *Nov. 2020 – May 2026*
Diploma (M.Sc.) in Physics; ***100/100 cum Laude*** — *Oct. 2018 – Sept. 2020*

University of Pisa — Pisa, Italy
M.Sc. in Condensed Matter Physics; ***110/110 cum Laude*** — *Oct. 2018 – Oct. 2020*

Scuola Superiore di Catania — Catania, Italy
Diploma (B.Sc.) in Physics; ***70/70 cum Laude*** — *Oct. 2015 – Sept. 2018*

University of Catania — Catania, Italy
B.Sc. in Physics; ***110/110 cum Laude*** — *Oct. 2015 – July 2018*

Liceo Scientifico "Leonardo da Vinci" — Reggio Calabria, Italy
High School Diploma; ***100/100 cum Laude*** — *Sept. 2010 – June 2015*

Teaching Experience

Forensic Department of the Italian Police — Caserta, Italy

- **Training Courses for Forensic Trainees:** Prepared and delivered 18 hours of lectures and hands-on training sessions on Electron Microscopy, Energy-Dispersive Spectroscopy, and standard forensic procedures for Gunshot Residue analysis; designed instructional materials and practical activities for trainees of the national *Forensic Inspectors* course. (Nov. 2025)

Scuola Normale Superiore — Pisa, Italy
Lecturer, Teaching Assistant and Examiner

- "**Physics School in Pisa**": Prepared and gave 32 hours of lectures about topics in Classical and Modern Physics and led the associated problem-solving sessions, graded 200 applicants per year through written and oral exams. (2019 → 2026, 24 selected high school students/year)
- "**University Outreach Courses**": Designed and delivered 12 hours of interactive lessons on Problem-Solving Techniques for 260 handpicked high school students; introduced university-level physics through active learning and everyday physics curiosities. (June - July 2024)
- **"Classical Mechanics and Thermodynamics" course:** Served as a teaching assistant, delivering 30 hours of problem-solving sessions for freshmen. (Fall 2020 - Spring 2021)

Italian Association for the Teaching of Physics Various locations across Italy
Lecturer, Coach and Examiner

- **"One step away from IPhO":** Prepared and delivered 40 hours of lectures on Celestial Mechanics, Electromagnetism, Thermodynamics, Optics and Special Relativity; developed and graded theoretical and experimental examinations for the selection of the Italian teams taking part in the International and European Physics Olympiads (IPhO and EuPhO). (May 2019/23/.../26, Italy's top 15 students/year)
- **Physics Olympiad Summer School:** Led 16 hours of problem-solving sessions on Orbital Mechanics, Estimation Problems, Dimensional Analysis and Fluid Dynamics, aimed at 20 handpicked students taking part in the Physics Olympiad. (Aug. 2019/24/25, ∼ 20 students/year)

Scuola Superiore di Catania Catania, Italy
Lecturer, Coach and Examiner

- **Physics Camp in Catania:** Prepared and gave 4 lectures about Electrostatics, Problem-Solving Techniques and Fluid Dynamics. Led the associated problem-solving sessions, prepared and graded the final theoretical competition. (Mar. 2023/.../26, ∼ 25 students/year)
- **Training camp for the Math Olympiad:** Led two problem-solving sessions focused on Combinatorics and Game Theory, aimed at 25 handpicked high school students taking part in the national finals of the Math Olympiad. (Apr. 2016 and 2017, ∼ 30 students/year)

Physics Education Projects

Italian Physics Olympiad (2019 – present) | *Italian website*

- Trainer of the Italian national teams for the European and International Physics Olympiads (EuPhO and IPhO) since 2019; member of the national committee since 2024, contributing to the design of theoretical and experimental problems and to multi-stage assessment and selection procedures involving an annual pool of approximately 30,000 participants. Students I have trained have earned 2 gold, 21 silver and 28 bronze medals at IPhO and EuPhO.

Physics School in Pisa (2019 – present) | *Italian website*

- Contributed to an annual one-week school for the top-performing Italian high school students in Physics. My main duties have included assessing about 200 applicants per year through written and oral exams, preparing original exam problems, and delivering lectures and problem-solving sessions. The project won the "Didactics of Physics" award from the Italian Physical Society.

Team Physics Competition (2023 – present) | *Italian website*

- Co-created a multi-stage team competition involving approximately three thousand participants annually; responsible for authoring, refining, and selecting theoretical problems and for ensuring consistent difficulty calibration.

Physics Camp in Catania (2023 – 2026)

- Member of the organizing team for a training camp combining lectures, problem-solving sessions, lab activities, and a concluding competition; supported curriculum planning and assessment.

Danny & Tony's Lab (2025 – present)

- Co-developing a fully custom-built virtual and interactive physics laboratory hosted on a website. The platform will allow users to perform a variety of physics experiments within a 3D environment where they can walk around and interact with the apparatus, as in a video game. The project aims to serve as an innovative and highly engaging tool for physics education.

"Problem-Solving Techniques in Physics, with Original Problems and Debunked Myths"

- Book aimed at Physics Olympiad and first-year university students; focuses on problem-solving strategies and systematic identification of misconceptions often found in traditional teaching and textbooks. Its Italian version is scheduled to be published in Feb. 2027.

Publications

Physics Education

D. B. P., Andrea Stefanini, Unblowing bubbles: understanding the physics of bubble deflation through a straw, Am. J. Phys. **93**, 797–805 (2025).

Stefania Carletti and D. B. P., *The Ninth Edition of the European Physics Olympiad*, chapter in La Fisica nella Scuola: Speciale Campionati di Fisica 2025, published by *Associazione per l'Insegnamento della Fisica* (2025, in Italian).

D. B. P., Debunking Boyle's self-flowing flask, Eur. J. Phys. **47** 015007 (2026).

D. B. P., The physics behind Heron's fountain, Am. J. Phys. **94**, 530–539 (2026).

D. B. P., *A rigorous derivation of the ideal gas law from energy constraints*, to be published on the Am. J. Phys. in August 2026.

D. B. P., *A tutorial on Bernoulli's equation, with frequent misconceptions and original examples*, in preparation.

Photonics & condensed-matter

D. B. P. and Giuseppe C. La Rocca, Interface mode between gyroelectric and hyperbolic media, J. Opt. Soc. Am. B **40**, 172 (2023).

D. B. P. and Giuseppe C. La Rocca, Bulk and surface modes in a one-dimensional gyro-uniaxial photonic crystal, Phys. Rev. A **110**, 033514 (2024).

D. B. P. and Z. Bacciconi, I. Carusotto, G. C. La Rocca, *Chiral quadrupolar optics of fractional quantum Hall liquids*, in preparation.

Conference Talks and Posters

Metamaterials'2025, Amsterdam (NL), Sept. 2025

Graviton-Polaritons in FQH Systems Coupled to Optical Cavities (P)

GIREP-EPEC Conference in Physics Education, Leiden (NL), July 2025

When Assumptions Make the Difference: The Curious Case of the Mistreated Bernoulli's Equation (T)

META 2024, Toyama (JP), July 2024

Photonic Modes in Gyrotropic-Hyperbolic Heterostructures (P)

Additional Courses

"GSR Analysis": Four-week intensive training program on the use of scanning electron microscopes and the application of X-ray, Raman, and FTIR spectroscopies for characterization of organic and inorganic gunshot residues. (Oct. 2025)

School for Forensic Police Inspectors: Selected through a national open competitive exam and completed a six-month training program in forensic science, including international law and experimental techniques used by the *Italian Forensics*, with a specialization in ballistics and GSR analysis. Appointed as *Forensic Inspector* upon completion. (June 2023 – Jan. 2024)

"24 ECTS for Teaching": Completed four prerequisite courses in Pedagogy, Psychology, Anthropology, and Teaching Methods, required for teacher qualification in Italy. (Summer 2022)

"English for Scientific Communication": Course on effective scientific writing and presentation, focusing on preparing research papers and delivering talks at international audiences. (Spring 2022)

Afternoon English Courses: Continuous English-language courses at a British School (grammar, writing, conversation) – 4 hours per week. Earned Cambridge English certifications up to C1. (Oct. 2008 – June 2015)

Awards & Achievements

Italy Team Leader at EuPhO 2025 and 2026: Coached and led the Italian team at the European Physics Olympiad, held in Sofia (Bulgaria, June 2025) and Göteborg (Sweden, June 2026).

Best student of the *Forensic Ballistics Inspector* course: Ranked first in the final examinations among the *ballistics inspector* trainees and third overall among all forensic trainees; awarded bronze medal. (Dec. 2023)

First Place in the National Open Competitive Exam for *Forensic Inspectors*: Ranked first in the national exam for *ballistic inspectors* to enroll in the Forensic Department of the Italian Police. (Apr. 2023)

Gold Medal – 53$^{\text{rd}}$ Rudolf Ortvay International Competition in Physics. (Feb. 2023)

Absolute Winner – 50$^{\text{th}}$ Rudolf Ortvay International Competition in Physics. (Dec. 2019)

SNS Scholarship for exceptional students: Awarded full scholarship (tuition waiver, free housing, meals and a monthly stipend) after passing the selection process to enroll at Scuola Normale Superiore as a Master's student. (Oct. 2018 – Sept. 2020)

SSC Scholarship: Granted full scholarship (room and board) after passing the selection process to enroll at Scuola Superiore di Catania as an undergraduate student. (Oct. 2015 – Sept. 2018)

Certifications and Skills

Languages: English (C1, 195/210, certified by Cambridge English), French (B1, certified by Scuola Normale Superiore).

Major computer skills: MATLAB, Mathematica, Python, LaTeX, Git, HTML, website management, basic video editing software.

Others: Fire Safety and First Aid (both certified by the Italian Police).